

Agam Katiyar

Bhopal, India

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TECHNICAL SKILLS

Programming Languages: Python, Java, SQL

AI/ML and Data: Scikit-learn, TensorFlow, OpenCV, MediaPipe, LangChain, Pandas, NumPy

Data Analysis and Visualization: Matplotlib, Seaborn, Exploratory Data Analysis (EDA)

Backend and APIs: REST APIs, FastAPI (basics), Docker (basics)

Databases: MySQL, MongoDB

Core CS and Tools: DSA, OOP, Operating Systems, Computer Networks, DBMS, Git, GitHub

WORK EXPERIENCE

Bittern Technologies

Noida

Artificial Intelligence Intern

May 2026 - June 2026

- Worked on Foliage Forensics, an automated plant disease classification project using machine learning techniques.
- Applied Python, Pandas, NumPy, and Scikit-learn for data preprocessing, model development, and evaluation.
- Assisted in implementing and evaluating supervised machine learning models for image-based classification tasks.
- Contributed to model experimentation using Decision Trees, SVM, PCA, and K-Means.

PROJECTS

Real-Time Speech-to-Speech Translator - [GitHub](#)

Tech Stack: Python, PyTorch, Whisper, NLLB-200, Tkinter

APR 2026

- Built a fully offline multilingual speech-to-speech translation system using OpenAI Whisper (STT) and Meta NLLB-200 (NMT) with a Tkinter GUI.
- Designed an end-to-end AI pipeline (ASR → NMT → TTS) with background threads to keep the GUI responsive during CPU-bound model inference.
- Resolved a Windows COM thread-safety crash by isolating TTS in a PowerShell subprocess and applied PyTorch inference optimizations.

Driver Drowsiness Detection System - [GitHub](#)

Tech Stack: Python, OpenCV, MediaPipe

MAR 2026

- Built a real-time driver monitoring system using MediaPipe to track 468 facial landmarks and detect drowsiness via EAR/MAR analysis at 25–30 FPS on CPU.
- Designed a temporal detection algorithm using consecutive-frame validation to reduce false alarms from natural eye blinks.
- Built an offline modular application with audio alerts and a live monitoring dashboard.

H-GestureControl: Hand Gesture Laptop Controller - [GitHub](#)

Tech Stack: Python, OpenCV, MediaPipe

JUL 2025

- Developed a real-time hand gesture recognition system using MediaPipe to classify 7 gestures from 21 hand landmarks at 30 FPS.
- Integrated 7 gesture-to-action mappings including volume control, media playback, screenshots, and screen locking using Windows Audio APIs, PyAutoGUI, and Win32 APIs.
- Improved gesture stability using cooldown timers, hysteresis, and an 18-frame rolling buffer to minimize false triggers.

EDUCATION

Vellore Institute of Technology, Bhopal

Bhopal

B.Tech in Computer Science Engineering (AI & ML)

AUG 2023 - AUG 2027

CGPA: 8.58/10

ACHIEVEMENTS AND CERTIFICATIONS

- GRABHACK 2025 Pre-Finalist – Ranked among Top 98 teams out of 4500+ teams nationwide
- Solved **400+ DSA problems** across LeetCode, GfG, and Codeforces
- ServiceNow University Virtual Internship Program – ServiceNow
- Applied Machine Learning in Python – University of Michigan(Coursera)
- Google AI Essentials – Google
- Earned Google Arcade swags through Google Cloud labs and quizzes
- NSS Event Volunteer – Contributed to the "10,000 Mega Tree Plantation in a Day" initiative